Neal Kaushik Sharma

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EDUCATION

Iowa State University — **Ames, Iowa**

B.S. in Computer Science Honors, Minor in Artificial Intelligence — **Expected: May 2028**

GPA: 3.8/4.0 - Dean's List for Academic Excellence — **Sophomore**

COMPUTATIONAL SKILLS

Languages: Java, Python, C, C++, C#, SQL, HTML, CSS, JavaScript

Tools and Frameworks: Git, Spring Boot, Flask, Django, .NET, Postman, SSH, Azure, LaTeX

EXPERIENCE

Software Engineer Intern – Tata Consultancy Services — **June 2025 - August 2025**

- Worked as a Web Application Developer, where my main project was BrewMaster (simulated coffee store), a full-stack online retail system built from scratch using C#, ASP.NET Core MVC, SQL Server, ADO.NET, jQuery, and AJAX (GitHub).
- Developed core system features, implementing secure authentication, image uploads, password hashing, role-based access, centralized logging, and a full MVC architecture connected to real-time SQL Server to handle inventory, orders, and product management efficiently.
- Rotated with frontend, backend, database, DevOps, QA, and cloud engineers, gaining exposure to the full software development lifecycle and seeing how their workflows and tools come together in an Agile team environment, while also learning practical skills from mentors such as API development, Bootstrap, Azure cloud, documentation, and presentation skills.

Cybersecurity & Software Development Assistant – ISEAGE — **January 2026 - Present**

- Support the planning and execution of five annual Cyber Defense Challenges (CDCs) by building and maintaining multi-application scenarios that simulate realistic operations, including web applications, services, and infrastructure, with hundreds of participants each competition.
- Manage Linux-based virtual machines hosting web applications, databases, and services; write scripts to deploy, start, stop, and reset services, configure networking between VMs, monitor resources, automate scenario setup, and integrate applications across multiple VMs.
- Debug software, fix broken services, and monitor systems during live competitions, responding to failures or attacks to keep scenarios running while teams defend against the penetration attempts of cybersecurity professionals.

Research - Malicious Unlearning Attacks against Predictive Uncertainty in AI Models — **August 2025 - Present**

- Research on how machine unlearning (data removal from pre-trained models) impacts predictive reliability in AI models, accepted for presentation at the *National Conference on Undergraduate Research (NCUR*).
- Implementing and comparing unlearning methods and data removal strategies on image classification models using real-world medical and benchmark datasets, focusing on model behavior, calibration, confidence shifts, Accuracy, and training time.
- Applying uncertainty estimation techniques (Deep Ensembles, Monte Carlo Dropout) and metrics such as Expected Calibration Error and Brier Score to quantify over-confidence or under-confidence of the predicted outputs and identify vulnerabilities introduced by unlearning.

PROJECTS

Space-Miner - 2D Java Game [GitHub] — **October 2025 - Present**

Personal project: A top-down survival/adventure game built entirely from scratch in Java, under active development with plans for public release.

- Developed core gameplay systems including precise player and object movement with collision detection, mining with particles and loot drops, companion AI (A* Search Algorithm), slot-based inventory, crafting, dynamic lighting, and day/night cycle.
- Built supporting systems such as multiple game states, tile-based worlds, minimap generation, screen culling, interactive UI with item descriptions and cursor controls, breakable objects, teleporting/transition triggers, and procedural object placement.
- Implemented save/load via Java serialization, integrated sound effects and pixel art assets, and added in-engine debugging tools.

Cy Saint's Hospital - Full-Stack Hospital Management App [GitHub] [Showcase] — **August 2025 - December 2025**

Developed as part of computer science coursework (COMS 3090) to streamline workflows for patients, doctors, pharmacists, and admins.

- Worked as a backend developer, gaining experience with Agile practices, coordinating roles, meeting deadlines, and applying technical design and teamwork to deliver a full-stack app. The project was recognized as *Best Project Overall (1st place) in a class of over 300 students*.
- Built key features including secure authentication, role-based access, appointment scheduling, prescription-driven pharmacy with protected medications, real-time messaging with Firebase powered notifications, video calling, and a privacy-focused AI health assistant.

Network Attached Storage — **June 2025 - October 2025**

Personal NAS and server on Raspberry Pi 5 with automated backups, SSH-based remote access, and secure networking.

- Configured OpenMediaVault to sync files via Syncthing across server and PCs, using version history and a custom 3D-printed enclosure.
- Integrated Tailscale for secure remote access, enabled local SMB network shares, implemented automated backups across multiple drives to prevent data loss, and deployed a Docker-based container environment for hosting web services.